

PYTHON DATA ANALYSIS PROJECT

Road Collision Severity Analysis in Great Britain

Exploratory Data Analysis using Python (2025 data)

INDUSTRY

Road Transport & Road Safety

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101,525

road collisions analysed in Great
Britain, 1 Jan to 31 Dec 2025

26,644

of them serious or fatal

Industry Overview: Road Transport & Safety



SELECTED INDUSTRY

Road Transport & Public Road Safety

Road transport moves people and goods across Great Britain. The Department for Transport, local authorities and police forces record every injury collision reported to the police, so they can see where, when and under what conditions collisions happen.

101,525

collisions recorded
in 2025

26,644

serious or fatal

365

days of data

Why data analysis matters here



Reveal hidden patterns

Shows which speed limits, road types and conditions go with serious outcomes.



Target limited resources

Directs enforcement, engineering and awareness work to the highest-risk times and places.



Support policy decisions

Gives evidence for speed-limit and road-design reviews instead of guesswork.



Prevent harm

Earlier detection of risk patterns helps reduce serious injuries and deaths.

Problem Statement



The Problem

- Collisions differ widely in severity and in the conditions around them.
- Raw records do not show which factors go with serious and fatal outcomes.
- Numeric codes and unlabelled date/time fields hide the patterns.



Why It Matters

- Serious and fatal collisions mean life-changing injury and loss of life.
- Enforcement and engineering budgets are limited and must be targeted.
- Knowing the risky conditions lets authorities act before more harm occurs.



Who Is Affected

- Road users and their families.
- Police forces and emergency services.
- Local authorities and national road-safety planners.
- Health services that treat the injured.

In 2025, 26.2% of recorded collisions (26,644 of 101,525) were serious or fatal.

Project Objectives

1

Prepare the data

Clean and transform the 2025 DfT collision records: missing values, duplicates, data types, readable labels and time features.

2

Profile severity

Measure the overall split of Slight, Serious and Fatal collisions.

3

Link conditions to severity

Compare severity across speed limit, area type, road type, junction, weather, road surface and vehicles involved.

4

Find when and where

Identify peak hours and days, and the local authority areas with the most collisions.

5

Summarise risk patterns

Find condition combinations common in serious and fatal collisions and turn them into insights.

Proposed Solution & Analysis Questions

Proposed solution

A reproducible Python EDA pipeline that turns the raw DfT collision file into labelled, analysis-ready data, then answers road-safety questions with tables and charts.

How Python helps

Pandas

clean, label, group, crosstab

NumPy

conditional features (np.select)

Matplotlib

bar and line charts

Seaborn

heatmaps, boxplots, density maps

Questions this project answers

Q1 What is the distribution of collision severity?

Q2 Which speed limits go with more severe collisions?

Q3 How do weather and road surface relate to severity?

Q4 Does severity differ between urban and rural areas?

Q5 At what times and days do collisions peak?

Q6 Which areas have the highest collision counts?

Dataset Information

101,525

Rows

44 / 59

Columns (raw / cleaned)

CSV

File format

2025

1 Jan to 31 Dec

DATASET NAME

Great Britain Road Collisions 2025 (cleaned file:
Great_Britain_Road_Collisions_2025_Cleaned.csv)

SOURCE

UK Department for Transport (DfT) road casualty statistics, collision
2025 file

DESCRIPTION

One row per police-reported injury collision, with location, road,
weather, vehicle, casualty and time attributes.

Important features

Target: collision_severity (1 Fatal, 2 Serious, 3 Slight)

Road: speed_limit, road_type, junction_detail

Environment: weather_conditions, road_surface_conditions,
light_conditions, urban_or_rural_area

Traffic & time: number_of_vehicles, number_of_casualties,
date, time, day_of_week

Location: latitude, longitude, local_authority_ons_district

Data Collection & Understanding

Collection: DfT 2025 collision CSV (police-reported injury collisions), loaded with `pd.read_csv`.

First look: `head()`, `shape`, `info()`, `describe()` and `dtypes` checked on a working copy.

Data types: 30 int64, 6 float64 and 8 text (str) columns in the raw 44.

Missing values: Only 2 rows (location fields: easting, northing, longitude, latitude).

Duplicates: 0 duplicate rows.

Needs attention: Coded categories (severity 1/2/3), dates and times stored as text, and -1 "missing or out of range" codes (e.g. 634 road-surface records).

```
In [ ]: cleaned_df.info()
```

```
RangeIndex: 101525 entries, 0 to 101524  
Data columns (total 44 columns):  
 3 location_easting_osgr    101523 non-null  
 4 location_northing_osgr  101523 non-null  
 5 longitude                101523 non-null  
 6 latitude                 101523 non-null  
 8 collision_severity       101525 non-null  
dtypes: float64(6), int64(30), str(8)
```

Notebook output: cleaned_df.info()

```
In [ ]: print("Duplicate rows:",  
            cleaned_df.duplicated().sum())  
mv = cleaned_df.isnull().sum()  
print(mv[mv > 0])
```

```
Duplicate rows: 0  
location_easting_osgr    2  
location_northing_osgr   2  
longitude                2  
latitude                 2
```

Notebook output: duplicate and missing value checks

Data Cleaning & Transformation

Working copy: raw_df copied to cleaned_df.

Missing values: 2 location rows filled with the median of their police force.

Duplicates: Checked: none to remove.

Data types: date converted to datetime; 0 invalid dates.

Inconsistent codes: Numeric codes mapped to labels (severity, day, road type, area, junction, weather, surface, light).

New columns: month, month_name, hour, time_period, vehicle_group (5+), local_authority_name.

Filtering: Urban/rural view drops 4 "Unallocated" rows; Serious + Fatal subset of 26,644 rows.

Result: 101,525 × 59 · 0 missing · 0 duplicates

```
In [ ]: for col in location_columns:
         cleaned_df[col] = (cleaned_df
                           .groupby("police_force")[col]
                           .transform(lambda x:
                                     x.fillna(x.median()))
         print(cleaned_df[location_columns]
               .isnull().sum())
```

location_easting_osgr	0
location_northing_osgr	0
longitude	0
latitude	0

Missing locations filled by police-force median

```
In [ ]: df["date_datetime"] = pd.to_datetime(
         df["date"], format="%d/%m/%Y")
         df["hour"] = df["time"].str.split(":")\
         .str[0].astype(int)
         df["time_period"] = np.select(
         conditions, choices, default="Unknown")
```

date	date_datetime	hour	time_period
04/02/2025	2025-02-04	15	Afternoon
07/05/2025	2025-05-07	9	Morning

Date/time conversion and new time features

Exploratory Data Analysis

Libraries & techniques

Pandas: value_counts, crosstab (normalised), groupby, map

NumPy: np.select, np.where for derived features

Seaborn / Matplotlib: charts (next slide)

Key findings

- Slight 73.76%, Serious 24.81%, Fatal 1.43%.
- 60 mph roads: highest serious + fatal rate, 36.38%.
- Rural 30.88% vs urban 23.87%.
- Peak hour is 17:00 (8,877 collisions); Friday is the busiest day.
- Mean casualties: Fatal 1.64, Serious 1.34, Slight 1.22.

Severity distribution

```
In [ ]: print(severity_summary)
```

	Severity	Collisions	Percentage
0	Slight	74881	73.76
1	Serious	25191	24.81
2	Fatal	1453	1.43

Serious + Fatal rate by speed limit (%)

```
In [ ]: print(severe_rate.round(2))
```

speed_limit	Serious_or_Fatal
20	22.35
30	24.81
40	27.27
50	29.58
60	36.38
70	25.54

Serious + Fatal rate by area type (%)

```
In [ ]: print(urban_rural_severe_rate.round(2))
```

Rural	30.88
Urban	23.87

Peak collision hour

```
In [ ]: print("Peak hour:", peak_hour)
        print("Collisions:", peak_hour_count)
```

Peak hour: 17
Collisions: 8877

Data Visualization

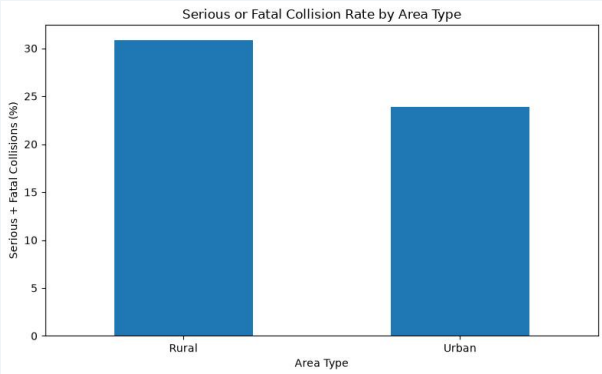


Line: Serious + Fatal rate by speed limit

Purpose: Compare severity risk across speed limits.

Shows: Serious or fatal share at 20 to 70 mph.

Insight: 22.4% at 20 mph rising to 36.4% at 60 mph, then 25.5% at 70 mph.

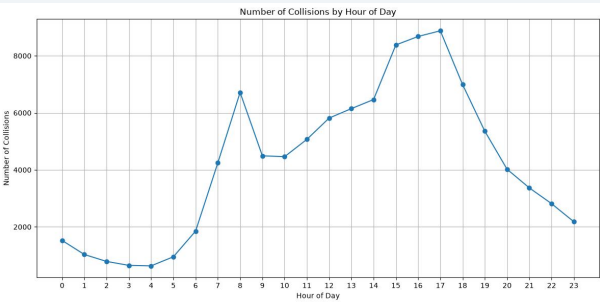


Bar: Serious + Fatal rate by area type

Purpose: Test if severity differs, urban vs rural.

Shows: Serious + Fatal share in each area type.

Insight: Rural 30.9% vs urban 23.9%; fatal share 2.7% vs 0.8%.

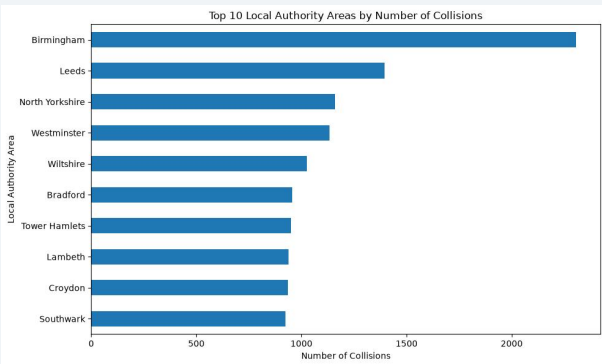


Line: Collisions by hour of day

Purpose: Find the busiest times of day.

Shows: Collision count per hour, 0 to 23.

Insight: Peak 17:00 (8,877); morning rise at 08:00 (6,716); lowest 04:00 (634).



Bar: Top 10 local authorities

Purpose: Locate areas with the most collisions.

Shows: Collisions in the ten busiest areas.

Insight: Birmingham leads (2,307), then Leeds (1,395); not adjusted for population or traffic.

Key Insights & Recommendations

KEY INSIGHT (from the data)

RECOMMENDATION

36.4%

Roads with a 60 mph limit have the highest serious + fatal rate (fatal 3.7%).

Review speed management and enforcement on 60 mph roads.

30.9%

Rural collisions are more often serious or fatal than urban (23.9%).

Prioritise rural road-safety measures.

73%

Single carriageways hold 74,150 of 101,525 collisions.

Focus engineering work on single carriageways.

17:00

Peak hour; Friday is the busiest day (16,811), Friday afternoon the busiest slot.

Schedule awareness campaigns and patrols around weekday afternoon peaks.

32.8%

Single-vehicle collisions are more often serious or fatal than two-vehicle ones (22.7%).

Investigate why single-vehicle collisions are more severe.

82%

Most collisions occur in fine weather (83,405); rare conditions like fog show higher severity but small samples.

Do not treat bad weather as the main driver; read rare categories with caution.

Conclusion



What was achieved

Cleaned and labelled 101,525 collision records, added time and area features, and answered the key severity questions with tables and charts.



Major findings

Severity is highest on 60 mph and rural roads; collisions peak at 17:00 and on Friday; single carriageways dominate.



How it addresses the problem

Shows authorities which conditions, times and places to target to reduce serious and fatal collisions.



Future scope

Build classification models to predict severity, add multi-year data, normalise by traffic or population, and test findings statistically.

Note: results show association, not cause. Categories with very small samples should be read with caution.